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24CS702 /22CS702
DATA ANALYTICS

Department: CSE/IT

Batch/Year: 2023-2027(CSE) & 2024-28(IT) / IV Year (CSE) & III Year (IT)

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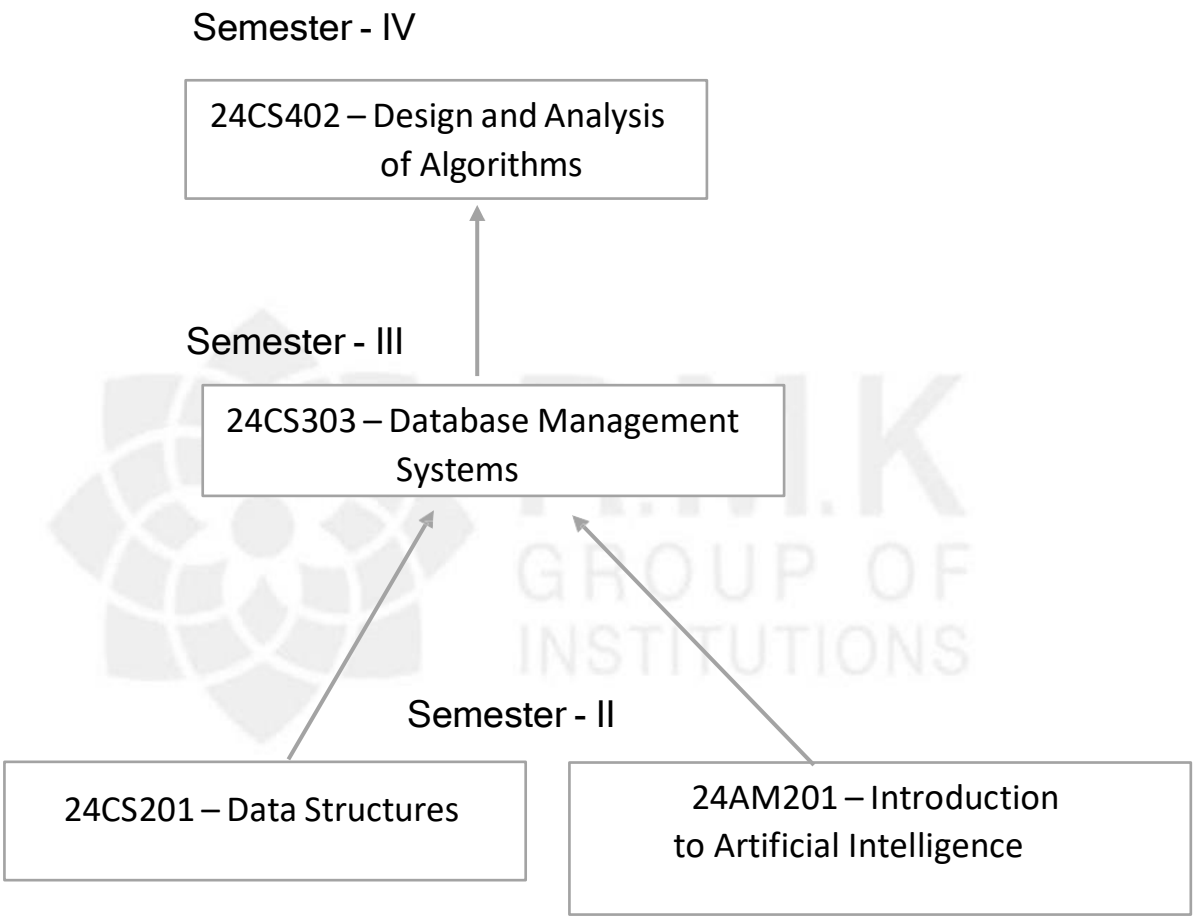
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2. Course Objectives

- To explain the fundamentals of big data and data analytics
- To discuss the Hadoop framework
- To explain about exploratory data analysis and data manipulation tools and use it for developing applications
- To analyze and interpret streaming data
- To discuss various applications of data analytics

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3. Pre-Requisites



4. SYLLABUS

24CS702 **DATA ANALYTICS** **L T P C**
3 0 2 4

UNIT I INTRODUCTION 9 + 6

Evolution of Big Data-Definition of Big Data-Challenges with Big Data- Traditional Business Intelligence (BI) versus Big Data-Introduction to big data analytics-Classification of Analytics-Analytics Tools- Importance of big data analytics.

UNIT II HADOOP FRAMEWORK 9+ 6

Introducing Hadoop- RDBMS versus Hadoop- Hadoop Overview-HDFS (Hadoop Distributed File System)- Processing Data with Hadoop- Managing Resources and Applications with Hadoop YARN - Interacting with Hadoop Ecosystem.

UNIT III EXPLORATORY DATA ANALYSIS 9+6

EDA fundamentals – Understanding data science – Significance of EDA – Making sense of data – Comparing EDA with classical and Bayesian analysis – Software tools for EDA –Data transformation techniques - Introduction to NoSQL – MongoDB: RDBMS Vs MongoDB – Data Types – Query Language – Hive – Hive Architecture – Data Types – File Formats - Hive Query Language (HQL) – RC File Implementation – User Defined Functions.

UNIT IV MINING DATA STREAMS 9+6

The data stream model – stream queries-sampling data in a stream-general streaming problem- filtering streams-analysis of filtering- dealing with infinite streams- Counting Distance Elements in a Stream – Estimating Moments – Counting Ones in Window – Decaying Windows.

UNIT V APPLICATIONS 9+6

Application: Sales and Marketing – Industry Specific Data Mining – microRNA Data Analysis Case Study – Credit Scoring Case Study – Data Mining Nontabular Data.

TOTAL: 75 PERIODS

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5. COURSE OUTCOMES

At the end of this course, the students will be able to:

COURSE OUTCOMES		KL
CO1	Explain the fundamentals of big data and data analytics and illustrate it.	K2
CO2	Discuss the Hadoop framework.	K2
CO3	Develop applications using exploratory data analysis and data manipulation tools.	K3
CO4	Analyze and interpret streaming data.	K4
CO5	Illustrate various applications of data analytics.	K2
CO6	Build applications for various domain.	K6

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CO – PO/PSO Mapping

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6. CO – PO /PSO Mapping Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS 03
CO1	3	2	1	2	2	-	-	1	2	1	-	3	3	3	3
CO2	3	2	1	2	3	-	-	1	2	2	-	2	3	3	3
CO3	3	2	2	3	3	-	-	1	2	1	2	3	3	3	3
CO4	3	3	2	3	3	-	-	1	2	1	-	3	3	3	3
CO5	3	3	2	3	1	-	-	1	2	2	2	3	3	3	3
CO6	3	3	3	3	2	-	-	1	2	2	2	3	3	3	3

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Lecture Plan

Unit - I

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LECTURE PLAN – Unit 1 - INTRODUCTION

Sl. No.	Topic	Number of Periods	Proposed Date	Actual Lecture Date	CO	Taxonomy Level	Mode of Delivery
1	Evolution of Big Data	1			1	K1	PPT/Chalk & Talk
2	Definition of Big Data	1			1	K2	PPT/Chalk & Talk
3	Challenges with Big data	1			1	K2	PPT/Chalk & Talk
4	Traditional Business Intelligence (BI) versus Big Data	2			1	K2	PPT/Chalk & Talk
5	Introduction to big data analytics	1			1	K2	PPT/Chalk & Talk
6	Classification of Analytics, Analytics Tools	2			1	K2	PPT/Chalk & Talk
7	Importance of Big data analytics	1			1	K2	PPT/Chalk & Talk
8	Installation of Python	3			1	K2	PPT
9	Working with Numpy Arrays	3			1	K2	PPT

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8. ACTIVITY BASED LEARNING

Activity name:

Analyzing Big Data Concepts through Social Media Platforms

Guidelines to do an activity:

1. Students can form groups (3 students per team)
2. Choose any one social media platform
   (e.g., Instagram / YouTube / Twitter / Facebook)
3. Analyze the platform based on:
   - Type of data generated
   - How data is collected and processed
   - How recommendations (feeds, reels, ads) are generated
4. Compare Traditional BI vs Big Data in the selected platform
5. Identify challenges and suggest possible solutions
6. Prepare a case study or presentation (charts, diagrams, examples)
7. Conduct Peer Review
   (Each team will be reviewed by other teams and mentors)Students can form group.

Useful link:

https://www.educba.com/business-intelligence-vs-big-data

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UNIT I
INTRODUCTION

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9. LECTURE NOTES

1. Evolution of Big Data

1970s and before was the era of mainframes. The data was essentially primitive and structured. Relational databases evolved in 1980s and 1990s. The era was of data intensive applications. The World Wide Web (WWW) and the Internet of Things (IOT) have led to an onslaught of structured, unstructured, and multimedia data.

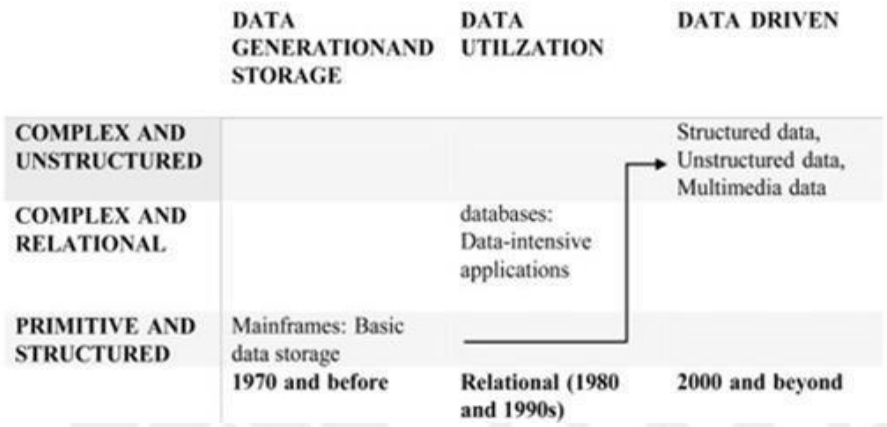


Fig.1:The evolution of big data

➢ **Mainframes** are high-performance computers with large amounts of memory and processors that process billions of simple calculations and transactions in real time.

Some examples of mainframe computer include:

- zSeries mainframe computer of Internation Business Machines series (IBM)
- Universal Automatic Computer from UNIVAC series.
- A system named NonStop by the manufacturers of Hewlett Packard.

➢ A **relational database** is a type of database that stores and provides access to data points that are related to one another.

- Examples of standard relational databases include Microsoft SQL Server, Oracle Database, MySQL and IBM DB2.

➢ **Data-intensive** is used to describe applications that are I/O bound or with a need to process large volumes of data. Such applications devote most of their processing time to I/O and movement and manipulation of data.

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- **Structured data** is highly organized and easily understood by machine language. Examples of structured data include names, dates, addresses, credit card numbers, stock information, geolocation, and more.
- **Unstructured data** is data stored in its native format and not processed until it is used. Examples of unstructured data includes things like video, audio or image files, as well as log files, sensor or social media posts.
- The **multimedia data** include one or more primary media data types such as text, images, graphic objects (including drawings, sketches and illustrations) animation sequences, audio and video.

2. Definition of Big Data

The following are the few responses for the question "Define big data" that we have heard over time.

1. Anything beyond the human and technical infrastructure needed to support storage, processing and analysis.
2. Today's BIG may be tomorrows NORMAL.
3. Terabytes or petabytes or zettabytes of data

All of these responses are correct. But it is not just one of these: in fact data is all of the above and more

Big Data refers to extremely large and complex datasets that cannot be managed, processed, or analyzed using traditional data-processing tools or software. It involves huge volumes of data that are continuously generated at high speed from various sources.

Core Concepts (The 5 Vs of Big Data)

1. Volume – Refers to the massive amount of data (from terabytes to zettabytes).
2. Velocity – The speed at which new data is generated and needs to be processed.
3. Variety – The different types of data: structured (tables), semi-structured (JSON, XML), and unstructured (videos, texts, images).
4. Veracity – The quality and trustworthiness of the data.

5. Value – The meaningful insights or business advantages derived from analyzing the data.

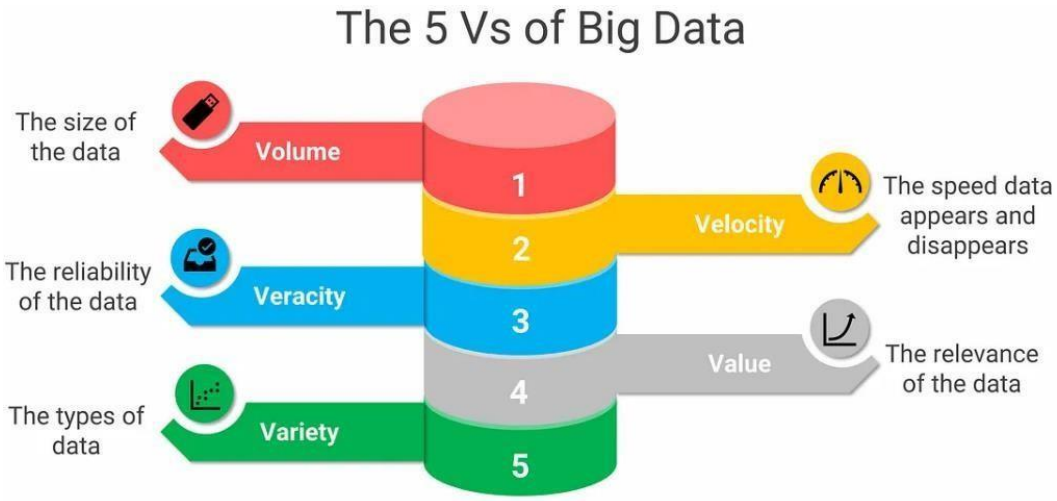


Fig.2:The 5 Vs of big data

Why Is Big Data Important?

- Helps businesses make data-driven decisions.
- Enables real-time insights into customer behavior.
- Powers AI/ML models, fraud detection, and predictive analytics.
- Supports innovation in sectors like healthcare, finance, retail, and transportation.

3. Challenges with big data

Following are a few challenges with big data:

1. Data today is growing at an exponential rate. Most of the data that we have today has been generated in the last 2-3 years. This high tide of data will continue to rise incessantly. The key questions here are: "Will all this data be useful for analysis?", "Do we work with all this data or a subset of it?", "How will we separate the knowledge from the noise?", etc.

2. Cloud computing and virtualization are here to stay. Cloud computing is the answer to managing infrastructure for big data as far as cost-efficiency, elasticity and easy upgrading/downgrading is concerned. This further complicates the decision to host big data solutions outside the enterprise.

3. The other challenge is to decide on the period of retention of big data. Just how long should one retain this data? A tricky question indeed as some data is useful for making long-term decisions, whereas in few cases, the data may quickly become irrelevant and obsolete just a few hours after having being generated.
4. There is a lack of skilled professionals who possess a high level of proficiency indata sciences that is vital in implementing big data solutions.

5. There are other challenges with respect to capture, storage, preparation, search, analysis, transfer, security and visualization of big data. Big data refers to datasets whose size is typically beyond the storage capacity of traditional database software tools. There is no explicit definition of how big the dataset should be for it to be considered "big data." Here we are to deal with data that is just too big, moves way to fast and does not fit the structures of typical database systems. The data changes are highly dynamic and therefore there is a need to ingest this as quickly as possible.

6. Data visualization is becoming popular as a separate discipline. We are short by quite a number, as far as business visualization experts are concerned.

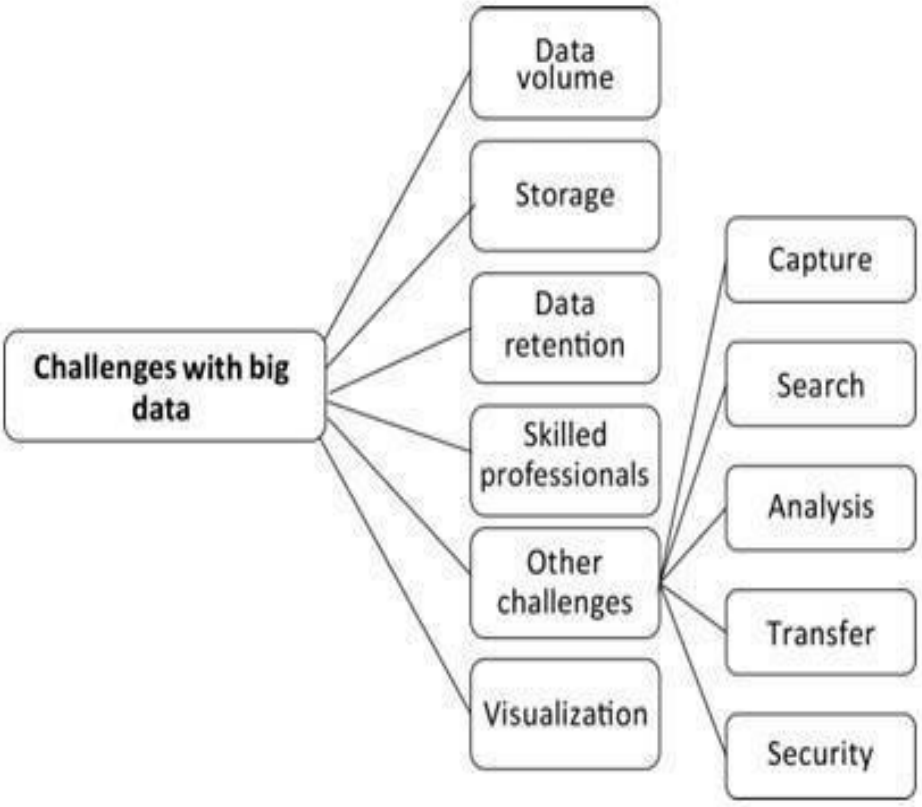


Fig.3:The Challenges with big data

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4. Traditional Business Intelligence (Bi) Versus Big Data

Following are the differences that one encounters dealing with traditional Business intelligence and big data.

1. In traditional BI environment, all the enterprise's data is housed in a central server whereas in a big data environment data resides in a distributed file system. The distributed file system scales by scaling in or out horizontally as compared to typical database server that scales vertically.
2. In traditional BI, data is generally analyzed in an offline mode whereas in big data, it is analyzed in both real time as well as in offline mode.
3. Traditional Bl is about structured data and it is here that data is taken to processing functions (move data to code) whereas big data is about variety: Structured, semi-structured and unstructured data and here the processing functions are taken to the data (move code to data).

5. Introduction to Big Data Analytics

Big Data Analytics is...

1. Technology-enabled analytics: Quite a few data analytics and visualization tools are available in the market today from leading vendors such as IBM, Tableau, SAS, R Analytics, Statistica, World Programming Systems (WPS), etc. to help process and analyze the big data.
2. About gaining a meaningful, deeper, and richer insight into the business to steer it in the right direction, understanding the customer's demographics to cross-sell and up- sell to them, better leveraging the services of the vendors and suppliers, etc.
3. About a competitive edge over the competitors by enabling us with findings that allow quicker and better decision-making.
4. A tight handshake between three communities: IT, business users, and data scientists. Refer Figure 4.
5. Working with datasets whose volume and variety exceed the current storage and processing capabilities and infrastructure of the enterprise.
6. About moving code to data. This makes perfect sense as the program for distributed processing is tiny (just a few KBs) compared to the data (Terabytes or Petabytes today and likely to be Exabytes or Zettabytes in the near future).

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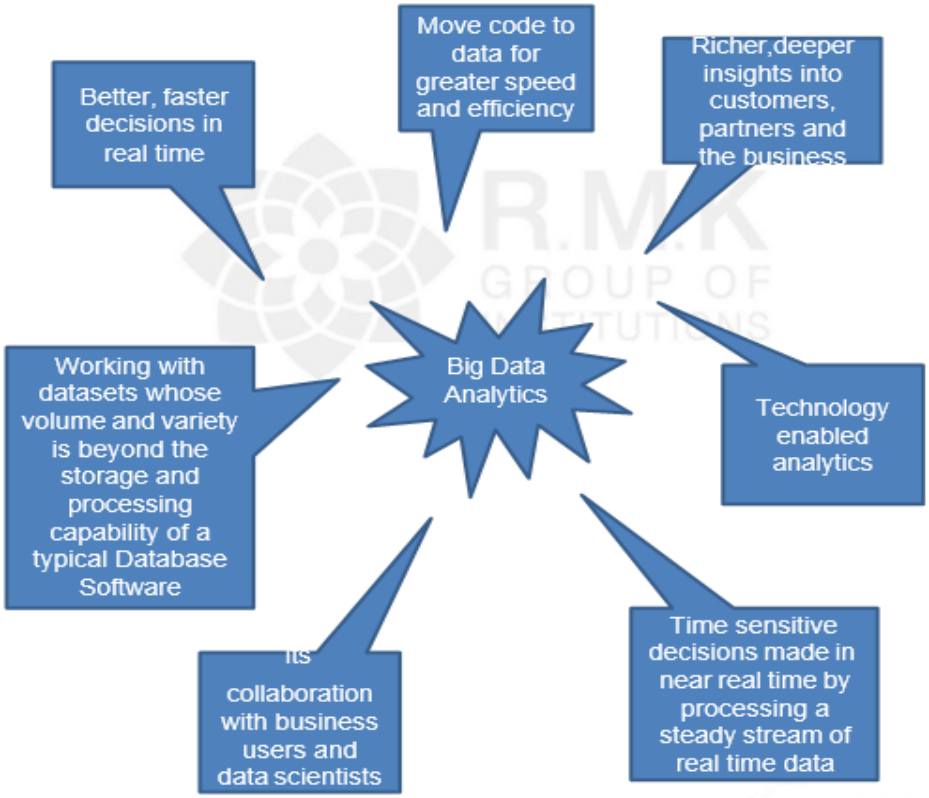


Fig.4: The big data Analytics

6. Classification of Analytics

There are basically two schools of thought:

1. Those that classify analytics into basic, operationalized, advanced and Monetized.
2. Those that classify analytics into analytics 1.0, analytics 2.0, and analytic 3.0.

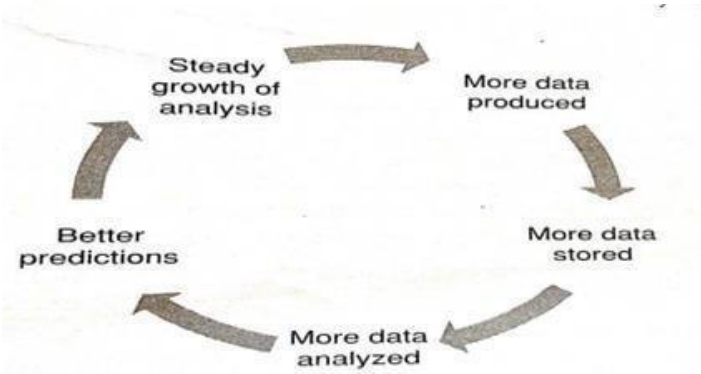


Fig.5: Big data entails

6.1 First School of Thought

It includes Basic analytics, Operationalized analytics, Advanced analytics and Monetized analytics.

- Basic analytics: This primarily is slicing and dicing of data to help with basic business insights. This is about reporting on historical data, basic visualization, etc.
- Operationalized analytics: It is operationalized analytics if it gets woven into the enterprise's business processes.
- Advanced analytics: This largely is about forecasting for the future by way of predictive and prescriptive modelling.
- Monetized analytics: This is analytics in use to derive direct business revenue.

6.2 Second School of Thought

Let us take a closer look at analytics 1.0, analytics 2.0, and analytics 3.0. Refer Table 6.1.

Table 6.1 Analytics 1.0, 2.0 and 3.0

Analytics 1.0	**Analytics 2.0**	**Analytics 3.0**
Era: mid 1950s to 2009 Descriptive statistics (report on events, occurrences, etc. of the past)	**2005 to 2012** Descriptive statistics + predictive statistics (use data from the past to make predictions for the future)	**2012 to present** Descriptive + predictive + predictive statistics (use data from the past to make prophecies for the future and at the same time make recommendations to leverage the situation to one's advantage)
key questions asked: What happened? Why did it happen?	key questions asked: What will happen? Why will it happen?	Key questions asked: What will happen? When will it happen? Why will it happen? What should be the action taken to take advantage of what will happen?
Data from legacy systems, ERP, CRM, and 3rd party applications.	Big data	A blend of big data and data from legacy systems, ERP, CRM, and 3rd party applications.

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Small and structured data sources. Data stored in enterprise data warehouses or data marts.	Big data is being taken up seriously. Data is mainly unstructured, arriving at a much higher pace. This fast flow of data entailed that the influx of big volume data had to be stored and processed rapidly, often on massive parallel servers running Hadoop.	A blend of big data and traditional analytics to yield insights and offerings with speed and impact.
Data was internally sourced.	Data was often externally sourced.	Data is both being internally and externally sourced.
Relational databases	Database appliances, Hadoop clusters, SQL to Hadoop environments, etc.	In memory analytics, in database processing, agile analytical methods, machine learning techniques, etc.

The **Classification of Analytics** is generally divided into **four main types**

1. **Descriptive Analytics – *"What happened?"***
   - **Purpose**: Summarize historical data to identify patterns or trends.
   - **Techniques**: Dashboards, reports, data aggregation
   - **Examples**:
     - Monthly sales reports
     - Web traffic statistics
     - Average customer satisfaction score

2. **Diagnostic Analytics – *"Why did it happen?"***
   - **Purpose**: Find the root cause of a problem or event.
   - **Techniques**: Drill-down, data discovery, correlations
   - **Examples**:
     - Sales dropped in March — why?
     - Bounce rate increased — is it due to page load time?
     - Analyzing low product ratings to identify design issues

3. **Predictive Analytics – *"What is likely to happen?"***
   - **Purpose**: Use historical data to forecast future outcomes.

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- **Techniques**: Statistical modeling, machine learning
- **Examples**:
    - Predicting customer churn
    - Forecasting inventory needs
    - Estimating demand during holidays

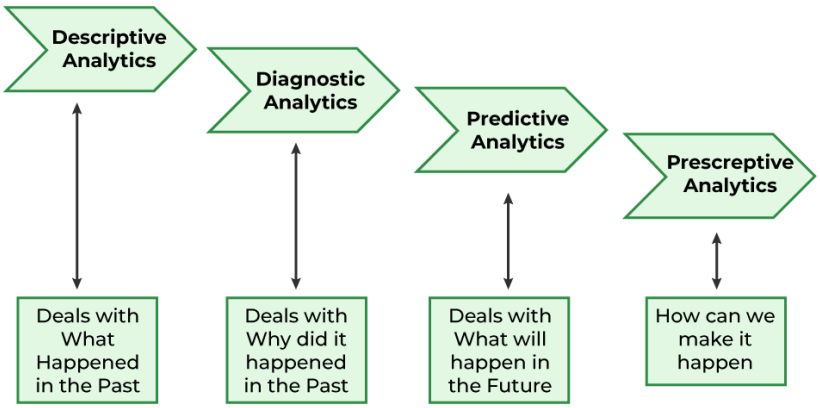


Fig.5: Types of cclassification of Analytics

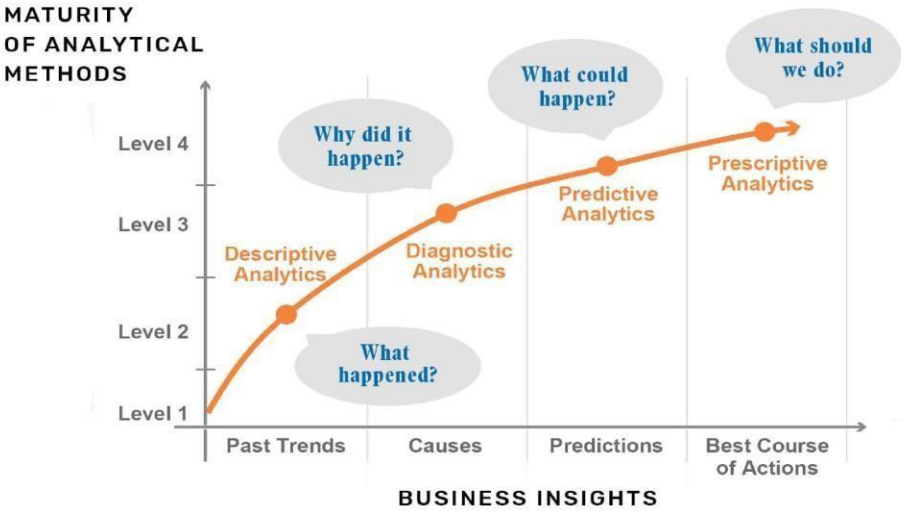


Fig.6: Classification of Analytics

4. Prescriptive Analytics – *"What should we do?"*

- **Purpose**: Recommend actions based on predictions and objectives.
- **Techniques**: Optimization, simulation, decision modelling

- **Examples**:
    - Recommending best shipping routes
    - Dynamic pricing models
    - Automated marketing strategies

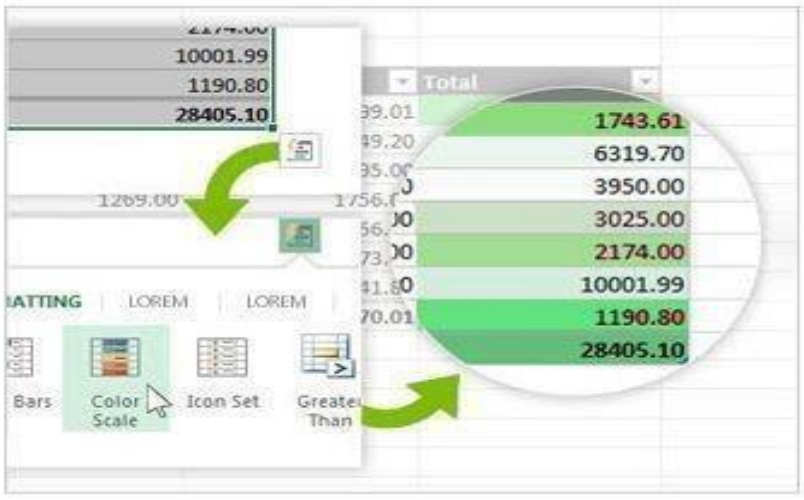
7. Analytics Tools

1. MS Excel

Templates do most of the set-up and design work for us, so we can focus on our data. In Excel 2013, there are templates for budgets, calendars, forms, and reports, and more.

➢ Instant Data Analysis

The new **Quick Analysis** tool lets us to convert our data into a chart or table in two steps or less. We can preview our data with conditional formatting,



sparklines, or charts, and make our choice stick in just one click.

➢ Powerful data analysis

- Create a PivotTable that suits our data
- Use multiple tables in our data analysis
- Power Query
- Power Map
- Connect to new data sources
- Use OLAP calculated members and measures

2. SAS

SAS is **a tool for analyzing statistical data**. SAS is an acronym for statistical analytics software. The main purpose of SAS is to retrieve, report and analyze statistical data.

3. IBM SPSS Modeler

IBM SPSS Modeler is a data mining and text analytics software application from IBM. It is used **to build predictive models and conduct other analytic tasks**. It has a visual interface which allows users to leverage statistical and data mining algorithms without programming

4. Statistica

- It's a worldwide proven software suite that can be used to analyze and visualize data, to create insights and predictions, and deliver tailored solutions
- More than 1 million users worldwide from a wide variety of industries rely on Statistica.
- The software stands out for its user-friendliness including the complete visualization of analytical processes in the project interface.
- Statistica offers the capability to integrate with many other tools and platform. By integrating TIBCO Spotfire, we can take advantage of comprehensive dashboard functionality.
- By integrating R and/or Python we can make use of their extensive capabilities.

5. Salford systems (World Programming Systems)

- The Salford Predictive Modeler (SPM) software suite is a highly accurate and ultra-fast platform for developing predictive, descriptive, and analytical models.
- The SPM software suite's data mining technologies span classification, regression, survival analysis, missing value analysis, data binning and clustering /segmentation.
- SPM algorithms are considered to be essential in sophisticated data science circles.

6. WPS

- World Programming produces and sells WPS Analytics, an advanced analytics software platform for data engineers and data scientists.
- The platform offers interactive visual and programming tools for working with data, supporting languages of SAS, SQL, Python and R, and runs on workstation, server, cloud, and mainframe systems.
- Leading organisations around the globe in every industry depend on WPS analytics Software.

7.1 Open source Analytics Tools

Some of the most popular **open-source Big Data Analytics tools**, widely used for data processing, storage, analysis, and visualization:

1. Apache Hadoop

- **Purpose**: Distributed storage and batch processing
- **Key Features**: HDFS, MapReduce, scalable, fault-tolerant
- **Use Case**: Large-scale data storage and batch analytics

2. Apache Spark

- **Purpose**: Fast, in-memory data processing engine
- **Key Features**: Supports batch, real-time, streaming, ML
- **Use Case**: Real-time analytics, ML pipelines, graph processing

3. Apache Hive

- **Purpose**: SQL-like query engine for Hadoop
- **Key Features**: Query structured data using HQL
- **Use Case**: Data warehousing and analysis on Hadoop

4. Apache Pig

- **Purpose**: High-level platform for creating MapReduce programs
- **Key Features**: Pig Latin language, easy ETL tasks
- **Use Case**: Data preprocessing and transformation

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5. Apache Flink

- **Purpose**: Stream and batch data processing
- **Key Features**: True real-time processing, low latency
- **Use Case**: Real-time event analytics, fraud detection

6. Apache Mahout

- **Purpose**: Machine learning on big data
- **Key Features**: Scalable algorithms for clustering, classification
- **Use Case**: Recommendation systems, pattern recognition

7. Apache Kafka

- **Purpose**: Distributed event streaming platform
- **Key Features**: High throughput, fault-tolerant pub-sub system
- **Use Case**: Log aggregation, real-time analytics pipelines

8. ELK Stack (Elasticsearch, Logstash, Kibana)

- **Purpose**: Real-time search, logging, and visualization
- **Key Features**: Powerful dashboards, full-text search
- **Use Case**: Monitoring, security analytics, business intelligence

9. Druid

- **Purpose**: Real-time data ingestion and OLAP querying
- **Key Features**: High performance, low latency
- **Use Case**: Interactive dashboards and analytics at scale

10. Jupyter + Pandas + PySpark

- **Purpose**: Exploratory data analysis and interactive notebooks
- **Key Features**: Flexible, Python-based, great for prototyping
- **Use Case**: Data exploration, cleaning, and modeling

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8. Importance of Big Data Analytics

The various approaches to analysis of data and what it leads to are:

8.1 Reactive-Business Intelligence:

What does Business Intelligence (BI) help us with? It allows the businesses to make faster and better decisions by providing the right information to the right person at the right time in the right format. It is about analysis of the past or historical data and then displaying the findings of the analysis or reports in the form of enterprise dashboards, alerts, notifications, etc. It has support for both pre-specified reports as well as ad hoc querying.

8.2 Reactive - Big Data Analytics:

Here the analysis is done on huge datasets but the approach is still reactive as it is still based on static data.

8.3 Proactive - Analytics:

This is to support futuristic decision making by use of data mining predictive modelling, text mining, and statistical analysis. This analysis is not on big data as it still uses the traditional database management practices on big data and therefore has severe limitations on the storage capacity and the processing capability.

8.4 Proactive - Big Data Analytics:

This is filtering through terabytes, petabytes, exabytes of information to filter out the relevant data to analyze. This also includes high performance analytics to gain rapid insights from big data and the ability to solve complex problems using more data.

VIDEO LINKS

Unit – I

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10. VIDEO LINKS

Sl. No.	Topic	Video Link
1	Evolution of Big Data	https://www.youtube.com/watch?v=MTpByOSs5KI
2	Challenges with Big Data	https://www.youtube.com/watch?v=aPAmVReeUaE
3	Traditional Business Intelligence (BI) versus Big Data	https://www.youtube.com/watch?v=DinhsGbrydY
4	Introduction to big data analytics	https://www.youtube.com/watch?v=bY6ZzQmtOzk
5	Classification of Analytics	https://www.youtube.com/watch?v=-HVbMti4g8c
6	Analytics Tools	https://www.youtube.com/watch?v=P-bKqfKhqR8
7	Importance of big data analytics	https://www.youtube.com/watch?v=wMzVIN13wnQ

11. ASSIGNMENTS: UNIT – I

Assignment Questions:

1. Define Big Data and list its key characteristics. CO1 (K1)
2. Explain the evolution of Big Data and discuss the challenges associated with it. CO1 (k2)
3. Illustrate the classification of analytics with suitable real-time examples. CO1 (K3)
4. Compare and contrast Traditional Business Intelligence (BI) and Big Data based on components, tools, and characteristics. CO1 (K4)
5. Evaluate the importance of Big Data Analytics in modern applications and justify its impact in any one domain (e.g., healthcare, banking, social media). CO1 (K5)

Scenario based questions for Classification of Analytics:

Scenario 1: E-commerce Sales Report Analysis

Question:
An online shopping company analyzes its monthly sales data, identifies best-selling products, and summarizes revenue by region and category.

What analytics types are involved?

Answer Expected:
♣ **Descriptive Analytics** – for summarizing past sales data
♣ **Diagnostic Analytics** – for identifying reasons behind high or low sales

Scenario 2: Student Performance Prediction

Question:
A college uses students' attendance, internal marks, and previous exam scores to predict which students are likely to fail in the final exams.
Which class of analytics does this scenario represent?

Answer Expected:
♣ **Predictive Analytics** – It uses historical student data to forecast future performance

Scenario 3: Inventory Management Optimization

Question:
A retail store suggests how much stock should be ordered for the next month based on demand trends, seasonal variations, and current inventory levels.
What type of analytics is this?

Answer Expected:
♣ **Prescriptive Analytics** – It recommends actions (how much stock to order)

Scenario 4: Fraud Detection in Banking

Question:
A bank monitors real-time transactions and flags unusual patterns such as large withdrawals or transactions from unknown locations to prevent fraud.
Which analytics types are involved?

Answer Expected:
♣ **Predictive Analytics** – to identify suspicious patterns
♣ **Prescriptive Analytics** – to trigger alerts or block transactions

Scenario 5: Healthcare Diagnosis Analysis

Question:
A hospital analyzes patient records and lab reports to determine the root cause of a disease outbreak in a specific area.
What type of analytics is this?

Answer Expected:
♣ **Diagnostic Analytics** – to identify the cause of the problem

PART-A Q&A

UNIT-I

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12. PART A QUESTIONS: UNIT – I

1. Define big data. (CO1 , K1)

Big Data refers to extremely large and complex datasets that are difficult to store, manage, and process using traditional data processing tools and techniques.

Example:

- Social media platforms generate terabytes of user data every day—this qualifies as Big Data.

2. What are the types of data? (CO1 , K1)

- Structured Data – Data organized in rows and columns, easily stored in relational databases (e.g., SQL tables).
- Unstructured Data – Data without a predefined format, such as images, videos, social media posts, or emails.
- Semi-structured Data – Data that doesn't fit into traditional databases but has some organizational properties, like XML, JSON, or NoSQL databases.

3. List few data analytics tools. (CO1 , K1)

- Hadoop – For distributed storage and processing of large datasets.
- Spark – For fast, in-memory big data processing.
- Tableau – For data visualization and reporting.
- R and Python – For statistical analysis and machine learning.
- Power BI – For interactive data visualization and business intelligence.

4. What are the advantages of Big data? (CO1,K1)

- Improved Decision Making – Analyzing large datasets helps organizations make data-driven and accurate decisions.
- Enhanced Customer Experience – Big data allows businesses to understand customer behavior and personalize services.

5. What is the difference between basic and advanced analytics? (CO1,K1)

- Basic Analytics involves summarizing historical data using techniques like reporting, dashboards, and descriptive statistics (e.g., averages, totals).
- Advanced Analytics uses complex techniques like predictive modeling, machine learning, and statistical algorithms to forecast future trends or uncover hidden patterns.
- In short, basic analytics tells what happened, while advanced analytics explains why it happened or what is likely to happen.

6. What do you know about the term "Big Data"? (CO1, K1)

Big Data refers to large and complex datasets that cannot be processed using traditional data management tools. It is defined by the 3 Vs:

- Volume (huge amount of data),
- Velocity (speed of data generation), and
- Variety (different formats like text, images, videos).

It is used to gain insights, improve decision-making, and drive innovation.

7. What is big data analytics? (CO1, K1)

- Big Data Analytics is the process of examining large and complex datasets to uncover hidden patterns, correlations, trends, and insights. It uses advanced tools and techniques like data mining, machine learning, and predictive analytics to support better decision-making and strategic planning.

8. What is the difference between basic and advanced analytics? (CO1, K1)

- Basic Analytics focuses on describing past data using simple tools like reports, summaries, and charts (e.g., what happened).
- Advanced Analytics uses statistical models, machine learning, and predictive techniques to forecast future events or understand deeper patterns (e.g., why it happened or what will happen).

9. Define Big data analytics. (CO1, k1)

- Big Data Analytics is the process of analyzing large, complex datasets (big data) to uncover hidden patterns, correlations, and insights. It uses advanced tools and techniques, such as machine learning, data mining, and predictive modeling, to drive informed decision-making and solve business problems.

10. How do you classify data analytics? (CO1, K1)

Data Analytics can be classified into **four main types**:

1. **Descriptive Analytics** – Analyzes historical data to understand what has happened (e.g., reports, dashboards).
2. **Diagnostic Analytics** – Investigates data to understand why something happened (e.g., root cause analysis).
3. **Predictive Analytics** – Uses statistical models and machine learning to predict future trends (e.g., forecasting).
4. **Prescriptive Analytics** – Recommends actions based on data analysis to optimize outcomes (e.g., optimization models, decision-making support).

11. What are the various approaches to analysis of data? (CO1, K1)

- Exploratory Data Analysis (EDA) – Involves summarizing and visualizing data to identify patterns, trends, and outliers before applying formal modeling techniques.

- Inferential Data Analysis – Uses statistical methods to make inferences or predictions about a population based on a sample of data.
- Predictive Data Analysis – Uses historical data and machine learning algorithms to predict future trends or behaviors.
- Causal Data Analysis – Focuses on understanding cause-and-effect relationships between variables (e.g., regression analysis).

12. What are the features of Powerful data analysis? (CO1, K1)

- Accuracy – The ability to provide correct and reliable insights by ensuring data quality and precise models.
- Scalability – The capability to handle large volumes of data and perform analysis efficiently as data grows.
- Real-time Processing – The ability to process and analyze data in real-time, providing immediate insights for decision-making.
- Advanced Techniques – Utilization of machine learning, statistical models, and data mining techniques to extract deep insights from complex datasets.

13. Tell us how big data and Hadoop are related to each other.

(CO1, K1)

- Big Data refers to large, complex datasets that are difficult to manage and analyze using traditional tools. Hadoop is an open-source framework designed to process and store Big Data efficiently across distributed systems. It provides tools for storage (HDFS) and processing (MapReduce), enabling organizations to handle vast amounts of data at scale. Thus, Hadoop is a key technology for managing and analyzing Big Data.

14. Explain the steps to be followed to deploy a Big Data solution.

(CO1, K2)

- Define Objectives and Requirements – Identify the goals of the Big Data solution, such as improving decision-making, predicting trends, or optimizing operations, and define the data sources and processing needs.
- Select the Right Tools and Technologies – Choose the appropriate Big Data technologies (e.g., Hadoop, Spark, NoSQL databases) for data storage, processing, and analysis based on the requirements.
- Data Collection and Preparation – Gather data from various sources and preprocess it for analysis, ensuring data quality, consistency, and transformation.
- Design the Architecture – Create a scalable and flexible system architecture, typically involving distributed storage (e.g., HDFS) and processing frameworks (e.g., MapReduce, Spark).
- Implement and Test – Deploy the solution in a test environment, validate its performance, and ensure it meets the business requirements.
- Monitor and Optimize – Continuously monitor the system for performance issues and optimize for efficiency, reliability, and scalability.

15. What are the two open-source analytics tools? (CO1, K1)

- **Apache Hadoop** – A framework that allows for distributed storage and processing of large datasets, widely used for Big Data analytics.
- **R –** A programming language and software environment for statistical computing and graphics, commonly used for data analysis and machine learning.

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13. PART B QUESTIONS: UNIT – I

1. Apply the concept of Big Data evolution to explain how modern social media platforms handle large-scale data. (CO1, K3)
2. Demonstrate how various challenges of Big Data can be addressed in a real-time application such as e-commerce or healthcare. (CO1, K3)
3. Apply the fundamentals of data analytics to analyze a given dataset and extract meaningful insights. (CO1, K3)
4. Compare and analyze the different types of analytics and evaluate their effectiveness in solving real-world problems. (CO1, K4)
5. Analyze various analytics tools and differentiate them based on features, performance, and applications. (CO1, K4)
6. Examine the importance of Big Data analytics and analyze its impact across different industries. (CO1, K4)

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14. SUPPORTIVE ONLINE CERTIFICATION COURSES

NPTEL : **Data Analytics with Python - Course**

Coursera : **Introduction to Data Science in Python | Coursera**

Udemy : **Top Data Science Courses Online - Updated [May 2025]**

edX : **Course | Introduction to Python for Data Science | edX**

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15. REAL TIME APPLICATIONS

The real-time applications of Big Data Analytics across various industries:

1. Healthcare

- **Remote Patient Monitoring**: Wearable devices send real-time vitals to healthcare providers.
- **Emergency Alerts**: Detect anomalies in patient data and trigger alerts for immediate response.
- **Disease Outbreak Prediction**: Analyzing real-time health reports and travel data.

2. Finance

- **Fraud Detection**: Real-time tracking of transactions to identify suspicious activity.
- **High-Frequency Trading**: Algorithms make split-second trading decisions using real-time market data.
- **Credit Scoring**: Dynamic analysis of financial behavior and risk profiles.

3. Retail & E-commerce

- **Personalized Marketing**: Real-time recommendation engines (e.g., "Customers also bought…").
- **Inventory Management**: Adjusting stock levels and supply chain based on live demand data.
- **Dynamic Pricing**: Adjusting prices based on current demand, competitor pricing, and customer behavior.

4. Transportation & Logistics

- **Fleet Tracking**: Real-time GPS data for managing logistics and delivery routes.
- **Predictive Maintenance**: Sensors predict part failures to avoid breakdowns.
- **Smart Traffic Systems**: Adjusting signals based on traffic flow data.

5. Telecommunications

- **Network Optimization**: Monitoring and resolving performance issues in real time.
- **Churn Prediction**: Identifying users likely to leave and offering immediate incentives.

6. Social Media & Entertainment

- **Trend Analysis**: Detect trending topics as they emerge.
- **Content Recommendation**: Real-time suggestions on platforms like Netflix, YouTube, Spotify.

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7. Manufacturing & Industry (IIoT)

- **Smart Factories**: Real-time analytics on production lines for efficiency.
- **Quality Control**: Immediate defect detection using sensors and AI.

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16. SELF LEARNING CONTENT: UNIT – I

Apache Pig: Overview

- Apache Pig is a high-level platform for creating programs that run on Apache Hadoop. It uses a language called Pig Latin, which abstracts the complexity of writing MapReduce programs.

Key Features of Apache Pig

- Simplified Data Flow: Pig Latin scripts are easier to write and understand than raw MapReduce.
- Extensible: Users can create UDFs (User Defined Functions) in Java, Python, etc.
- Optimized Execution: Pig handles optimization internally.
- Supports Structured & Semi-structured Data: Works well with nested structures like JSON.

Pig Latin Example

Step 1: Load Data

```
students = LOAD 'students.txt' USING PigStorage(',')
    AS (id:int, name:chararray, age:int, score:int);
```

Step 2: Filter Data

```
high_scorers = FILTER students BY score > 75;
```

Step 3: Group & Aggregate

```
grouped_by_age = GROUP students BY age;
avg_score = FOREACH grouped_by_age GENERATE group AS age,
AVG(students.score) AS avg_score;
```

Step 4: Sort Records

```
student_count = FOREACH (GROUP students ALL) GENERATE
 COUNT(students);
```

Step 5: Count Records

```
student_count = FOREACH (GROUP students ALL) GENERATE
COUNT(students);
```

Step 6: Join Datasets

```
grades = LOAD 'grades.txt' USING PigStorage(',') AS (id:int, grade:chararray);
joined_data = JOIN students BY id, grades BY id;
```

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Step 7: Store Result

```
STORE high_scorers INTO 'output/high_scores' USING PigStorage(',');
```

Common Use Cases

- Data preprocessing before machine learning.
- ETL operations (Extract, Transform, Load).
- Log analysis and transformation

Applications of Apache Pig:

1. Data Transformation (ETL)

- Apache Pig is widely used to **Extract, Transform, and Load (ETL)** data.
- Example: Clean and normalize log files before storing in HDFS or passing to a data warehouse.

2. Log Processing

- Used to process **large volumes of server logs** for analysis.
- Common in companies that need to monitor web traffic, user behavior, or error patterns.

3. Data Aggregation and Summarization

- Aggregate huge datasets (e.g., user clicks, transactions).
- Summarize data by calculating totals, averages, counts, etc.

4. Data Pipeline for Machine Learning

- Pig can preprocess raw data (cleaning, filtering, transforming) before feeding into ML models.
- Often paired with tools like Mahout or exported to Spark/MLlib.

5. Processing Semi-structured Data

- Pig supports data formats like **JSON**, **XML**, or nested structures, making it ideal for semi-structured sources.

6. Prototyping MapReduce Jobs

- Faster development than Java-based MapReduce.
- Great for quick prototyping and testing of data processing logic.

7. Integrating Data from Multiple Sources

- Combine data from multiple files, databases, or HDFS locations to produce unified datasets.

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8. Social Media and E-commerce Analytics

- Analyze customer reviews, clickstreams, and product performance in real-time or batch jobs.

Types of Data Models in Apache Pig:

It consist of the 4 types of data models

- **Atom**: It is a atomic data value which is used to store as a string. The main use of this model is that it can be used as a number and as well as a string.
- **Tuple**: It is an ordered set of the fields.
- **Bag**: It is a collection of the tuples.
- **Map**: It is a set of key/value pairs.

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17. Assessment Schedule (Proposed Date & Actual Date)

Sl. No	ASSESSMENT	Proposed Date	Actual Date
1	FIRST INTERNAL ASSESSMENT		
2	SECOND INTERNAL ASSESSMENT		
3	MODEL EXAMINATION		
4	END SEMESTER EXAMINATION		

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18. PRESCRIBED TEXT BOOKS & REFERENCE BOOKS

TEXT BOOKS:

1. Subhashini Chellappan, Seema Acharya, "Big Data and Analytics", 2nd edition, Wiley Publications, 2019.
2. Suresh Kumar Mukhiya and Usman Ahmed, "Hands-on Exploratory Data Analysis with Python", Packt publishing, March 2020.
3. Jure Leskovek, Anand Rajaraman and Jefrey Ullman," Mining of Massive Datasets. v2.1", Cambridge University Press,2019.
4. Glenn J. Myatt, Wayne P. Johnson, Making Sense of Data II : A Practical Guide To Data Visualization, Advanced Data Mining Methods, and Applications, Wiley 2009.

REFERENCES:

5. Nelli, F., Python Data Analytics: with Pandas, NumPy and Matplotlib, Apress, 2018.
6. Bart Baesens," Analytics in a Big Data World: The Essential Guide to Data Science and its Applications", John Wiley & Sons, 2014
7. Min Chen, Shiwen Mao, Yin Zhang, Victor CM Leung, Big Data: Related Technologies, Challenges and Future Prospects, Springer, 2014.
8. Michael Minelli, Michele Chambers, Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends", John Wiley & Sons, 2013.
9. Marcello Trovati, Richard Hill, Ashiq Anjum, Shao Ying Zhu, "Big Data Analytics and cloud computing – Theory, Algorithms and Applications", Springer International Publishing, 2016.

19. MINI PROJECT SUGGESTION

1. Fake News Detection in Social Media
- Tools: Python (NLTK, Scikit-learn), TensorFlow, MongoDB
- Goal: Classify news articles/posts as real or fake using text analysis.
- Learning: NLP, classification models, text preprocessing.

2. Smart Healthcare Disease Prediction System
- Tools: Python (Pandas, Scikit-learn), Flask, MySQL
- Goal: Predict possible diseases based on patient symptoms and medical history.
- Features: Symptom-based input, prediction dashboard.

3. Movie Recommendation System
- Tools: Python (Pandas, Surprise Library), Spark MLlib
- Goal: Recommend movies based on user preferences and ratings.
- Dataset: MovieLens dataset.
- Learning: Collaborative filtering, recommendation systems.

4. Credit Card Fraud Detection System
- Tools: Python, Spark MLlib, Hadoop
- Goal: Detect fraudulent transactions using anomaly detection techniques.
- Features: Real-time alerts, pattern recognition.

5. Weather Forecasting and Analysis System
- Tools: Python, PySpark, Tableau/Power BI
- Goal: Analyze historical weather data and predict future conditions.
- Dataset: Open weather datasets (Kaggle/data.gov).

6. Job Market Trend Analysis Platform
- Tools: Python, Web Scraping (BeautifulSoup), Power BI
- Goal: Analyze job postings to identify in-demand skills and roles.
- Learning: Data extraction, trend analysis, visualization.

7. Energy Consumption Prediction System

8. • Tools: Python, Spark, IoT datasets
- Goal: Predict electricity usage based on historical consumption patterns.
- Use Case: Smart homes, energy optimization.

9. Online Product Review Rating Prediction
- Tools: Python (NLP, Scikit-learn), TensorFlow
- Goal: Predict product ratings based on textual reviews.
- Learning: Sentiment analysis, regression/classification.

10. Smart Agriculture Crop Recommendation System
- Tools: Python, MLlib, IoT datasets
- Goal: Recommend suitable crops based on soil, weather, and environmental conditions.
- Learning: Predictive analytics, agriculture applications.

20. GATE QUESTIONS

1. A linear regression model predicts sales (Y) using advertising spend (X): Y = 2.5X + 10. What will be the predicted sales when X = 40?

A) 100
B) 110
C) 90
D) 85

Answer: B) 110
Solution: Y = 2.5 × 40 + 10 = 100 + 10 = 110

2. A dataset has a mean of 50 and a standard deviation of 5. What is the z-score of a value 60?

A) 1
B) 2
C) 1.5
D) 3

Answer: B) 2
Solution: Z = (X - μ) / σ = (60 - 50) / 5 = 2

3. You have a dataset with the following correlation matrix snippet:

- Income vs Age: 0.3
- Income vs Education: 0.8
- Income vs Experience: 0.6

Which feature has the strongest linear relationship with Income?

A) Age
B) Education
C) Experience
D) All equal

Answer: B) Education
Solution: Highest correlation value (0.8) is with Education.

4. In a binary classification, a model has: TP = 80, FP = 10, FN = 20, TN = 90. What is the precision?

A) 0.8
B) 0.9

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C) 0.75
D) 0.5

Answer: B) 0.89
Solution: Precision = TP / (TP + FP) = 80 / (80 + 10) = 80 / 90 ≈ 0.89

5. A logistic regression model gives a probability output of 0.72 for a binary outcome. Using a threshold of 0.7, what will be the predicted class?

A) 0
B) 1
C) Cannot be determined
D) Depends on accuracy

Answer: B) 1
Solution: Since 0.72 > 0.7 threshold, class = 1

6. You perform PCA and find that the first two components explain 85% of the variance. What does this imply?

A) 85% of data is missing
B) First two components are enough for visualization
C) You need more components
D) Model has high bias

Answer: B) First two components are enough for visualization
Solution: 85% explained variance suggests strong representation in reduced dimensions.

7. In a k-means clustering output with 3 clusters, the silhouette score is 0.85. What does this indicate?

A) Poor clustering
B) Too many clusters
C) Well-separated clusters
D) Data is non-clusterable

Answer: C) Well-separated clusters
Solution: Silhouette scores close to 1 suggest well-separated and cohesive clusters.

8. A confusion matrix has: TP=50, TN=40, FP=20, FN=10. What is the F1 Score?

A) 0.71
B) 0.75

C) 0.80
D) 0.85

Answer: B) 0.75
Solution:
Precision = 50 / (50 + 20) = 0.714
Recall = 50 / (50 + 10) = 0.833
F1 = 2 × (0.714 × 0.833) / (0.714 + 0.833) ≈ 0.768 → Closest to **0.75**

9. A time series model shows a seasonal pattern every 12 months. What is the frequency of the seasonality?

A) 1
B) 12
C) 6
D) 0.083

Answer: B) 12
Solution: Frequency = number of periods per season = 12 months

10. A dataset has a strong positive skew. Which measure of central tendency is most affected?

A) Mean
B) Median
C) Mode
D) None

Answer: A) Mean
Solution: The **mean** is pulled in the direction of the skew more than the median or mode.

11. (GATE 2014 – Set Theory / Data Analytics Basics)

Which of the following represents the number of elements in the union of two sets A and B?

A) |A| + |B|
B) |A| + |B| − |A ∩ B|
C) |A ∩ B|
D) |A| × |B|

Answer: **B) |A| + |B| − |A ∩ B|**
12. (GATE 2016 – DBMS)

Which of the following operations is used to retrieve specific data from a database?

A) INSERT
B) DELETE
C) SELECT
D) UPDATE

Answer: **C) SELECT**
13. (GATE 2018 – Data Mining Concept)

Association rule mining is primarily used for:

A) Classification
B) Clustering
C) Finding frequent itemsets
D) Regression

Answer: **C) Finding frequent itemsets**
14. (GATE 2020 – Algorithms / Data Processing)

Which of the following best describes a divide-and-conquer approach used in large-scale data processing?

A) Sequential execution
B) Parallel processing
C) Iterative looping
D) Recursion only

Answer: **B) Parallel processing**
15. (GATE 2017 – Probability / Predictive Analytics)

If P(A) = 0.6, P(B) = 0.5 and P(A ∩ B) = 0.3, find P(A ∪ B).

Answer:
= 0.6 + 0.5 − 0.3 = 0.8
16. (GATE 2015 – DBMS / Data Warehousing)

Which of the following is used for analytical processing rather than transaction processing?

A) OLTP
B) OLAP
C) FTP
D) HTTP

Answer: **B) OLAP**
17. (GATE 2019 – Machine Learning Basics)

Which of the following is a supervised learning technique?

A) Clustering
B) Association Rule Mining
C) Classification
D) Dimensionality Reduction

Answer: **C) Classification**

18. (GATE 2021 – Data Structures / Big Data Handling)

Which data structure is most suitable for handling large-scale data efficiently?

A) Array
B) Linked List
C) Hash Table
D) Stack

Answer: **C) Hash Table**

19. (GATE 2013 – DBMS)

Which of the following ensures data consistency in databases?

A) Atomicity
B) Consistency
C) Isolation
D) Durability

Answer: **B) Consistency**

20. (GATE 2022 – Data Analytics Concept)

A system that summarizes past data into reports is an example of:

A) Predictive Analytics
B) Prescriptive Analytics
C) Descriptive Analytics
D) Diagnostic Analytics

Answer: **C) Descriptive Analytics**

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21. AI based Applications

1. Intelligent Healthcare Systems

Artificial Intelligence combined with Big Data is widely used in healthcare to analyze large volumes of patient data. AI models process structured and unstructured data such as medical records, lab reports, and imaging data.

Applications:

- Disease prediction (e.g., diabetes, cancer)
- Medical image analysis (X-rays, MRI scans)
- Personalized treatment recommendations
- Patient risk prediction and early diagnosis

2. **Smart Financial Systems**

Big Data analytics helps financial institutions process millions of transactions in real time. AI algorithms identify patterns and anomalies for better decision-making.

Applications:

- Fraud detection using anomaly detection techniques
- Credit scoring using predictive analytics
- Algorithmic trading using real-time data
- Risk management and investment prediction

3. Intelligent Recommendation Systems

Retail companies use Big Data to analyze customer behavior, preferences, and purchase history. AI enhances customer experience through personalization.

Applications:

- Product recommendation systems
- Customer segmentation
- Demand forecasting
- Dynamic pricing strategies

4. Smart Urban Analytics

Big Data from sensors, cameras, and IoT devices is analyzed using AI to improve city management and infrastructure.

Applications:

- Traffic prediction and management
- Smart energy usage
- Waste management optimization
- Public safety monitoring

5. Intelligent Transport Systems

AI analyzes real-time transportation data to improve efficiency, safety, and route optimization.

Applications:

- Route optimization using predictive analytics
- Autonomous vehicles
- Fleet management
- Traffic congestion prediction

6. Smart Manufacturing (Industry 4.0)

Big Data analytics helps industries monitor machines and production processes. AI ensures efficiency and reduces downtime.

Applications:

- Predictive maintenance
- Quality control using computer vision
- Process optimization
- Supply chain analytics

Thank you

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